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X-ray Tube Spectrum Simulation with PENELOPE

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X-ray Tube Spectrum Simulation

1. Methods and Setup

The aim of this practice is to study the photon spectrum emerging from an X-ray tube simulated with the Monte Carlo code PENELOPE, taking as starting point *example 5* from the `penmain` distribution. The geometry consists of a tilted anode bombarded by a monoenergetic electron beam, followed by an aluminium added filter and a silicon photon detector. Starting from the original configuration (tungsten anode, 45° tilt, 1 mm Al filter, $E_0 = 150$ keV), six simulations are produced by varying the electron energy ($150 \rightarrow 28$ keV), the anode material ($W \rightarrow Mo$ at 28 keV), and the anode angle and added filtration ($45^\circ/1 \text{ mm} \rightarrow 20^\circ/2 \text{ mm}$) for each of the three configurations above. The PENELOPE photon spectra are then compared with the corresponding analytical spectra produced by the `SPEKPY v2` toolkit in the same conditions, used here as a fast semi-empirical reference to cross-check the Monte Carlo results.

1.1. PENELOPE simulation

The simulation uses `penmain` from the PENELOPE-2018 distribution. The geometry of *example 5* is reused unchanged in its qualitative structure: a cylindrical anode body limited by a tilted plane defines the target; below it, a thin Al disc acts as the inherent plus added filter; further downstream a silicon disc records the photon spectrum as an impact detector. The relevant input variables modified across the six runs are summarised in Table 1; the geometry change is implemented by the THETA angle of the limiting plane in the `tube.geo` file (from -135° for a 45° anode to -160° for a 20° anode) and by the Z -coordinate of the lower face of the filter ($-5.1 \text{ cm} \rightarrow -5.2 \text{ cm}$ to double the Al thickness).

In all cases bremsstrahlung and characteristic-X-ray splitting were kept at the original splitting factor of 4, and interaction forcing was applied to reduce variance for photon production. For the low-energy (28 keV) cases the absorption energies were lowered to 1 keV (from the default 5 keV) to properly track the characteristic K-lines of Mo that lie below 20 keV, as well as the soft tail of the bremsstrahlung continuum.

Case	E_0 (keV)	Anode	Angle	Al filter
1 (baseline)	150	W	45°	1 mm
2	28	W	45°	1 mm
3	28	Mo	45°	1 mm
4	150	W	20°	2 mm
5	28	W	20°	2 mm
6	28	Mo	20°	2 mm

Table 1: Summary of the six PENELOPE simulations performed.

1.2. Analytical reference — SpekPy

The same configurations were reproduced with the open-source toolkit SPEKPY v2. SpekPy generates photon spectra at the tube exit by combining a semi-empirical bremsstrahlung model and tabulated characteristic-line emission, with optional filtering. The fluence option (`flu=True`) was used so that both spectra represent the photon fluence per unit energy and per primary electron, and both are area-normalised to unity to allow shape comparison on equal footing.

2. Original Spectrum (W, 150 keV, 45°, 1 mm Al)

The PENELOPE output for the original configuration is shown in Figure 1.

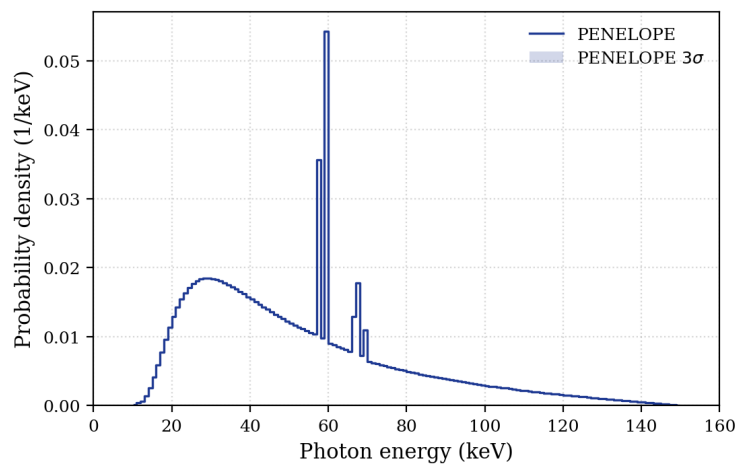


Figure 1: PENELOPE photon spectrum from the original *example 5* input (W anode, 45°, 1 mm Al filter, $E_0 = 150$ keV). Statistical uncertainty at 3σ is shown as a shaded band but is too narrow to be visible at this scale.

The spectrum displays the two classical ingredients of an X-ray tube emission: a continuous *bremsstrahlung* contribution extending from approximately 10 keV up to the endpoint at $E_0 = 150$ keV, with a maximum near 30 keV, and the W characteristic K-lines, clearly visible as the sharp peaks at $E_{K\alpha} \simeq 58\text{--}60$ keV and $E_{K\beta} \simeq 67\text{--}70$ keV. The low-energy cut-off near 10 keV is produced by the 1 mm Al filter, whose attenuation coefficient grows steeply below this energy; the high-energy edge at 150 keV is the kinematic limit set by the incident electron energy, since a single electron cannot produce a bremsstrahlung photon more energetic than itself. The very small 3σ band confirms that the Monte Carlo statistics are well converged.

The same configuration was reproduced with SpekPy and is overlaid on the PENELOPE spectrum in Figure 2. The two codes agree on the position of the W K-lines and on the overall shape of the bremsstrahlung continuum; the only systematic differences observed — and which will recur in all the following comparisons — are:

- *peak amplitudes*: small differences in K-line peak height, consistent with different tabulations of the K-shell fluorescence yields used by the two codes;
- *energy resolution*: SpekPy returns a continuous analytical curve while PENELOPE produces a histogram with finite bin width, which broadens the K-lines and lowers their apparent peak height even when the integrated intensity matches;
- *low-energy tail*: small deviations at the lowest energies, in the region most sensitive to the transport cut-offs of PENELOPE.

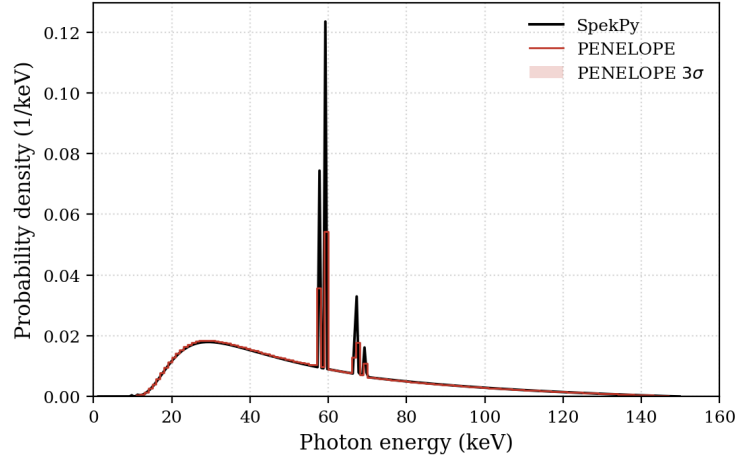


Figure 2: Comparison between the area-normalised PENELOPE photon spectrum (red, with the 3σ statistical band) and the SpekPy v2 analytical prediction (black) for the baseline case.

3. Effect of Beam Energy ($150 \rightarrow 28$ keV)

Reducing E_0 from 150 to 28 keV (Figure 3) produces three qualitatively new features.

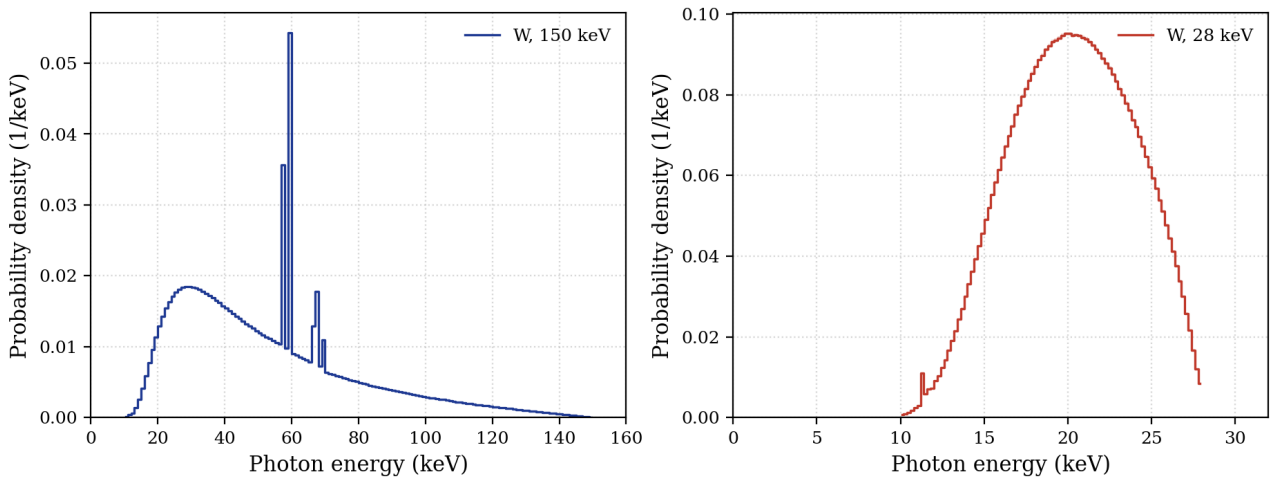


Figure 3: Effect of changing the electron beam energy at fixed geometry (W anode, 45° , 1 mm Al). *Left:* $E_0 = 150$ keV (baseline). *Right:* $E_0 = 28$ keV (mammographic regime).

First, the kinematic endpoint moves down to 28 keV, compressing the whole emission into the diagnostic mammographic window. Second, the W K-lines disappear entirely: the K-shell binding energy of W is ~ 69.5 keV, far above the available kinetic energy of the impinging electrons, so only L- and M-line emission can be excited, and these lines fall in the heavily absorbed $\lesssim 12$ keV region. Third, the Al filter, whose attenuation length increases steeply at low energy, now removes a much larger fraction of the spectrum: the bremsstrahlung continuum below ~ 10 keV is essentially wiped out, and the surviving distribution is concentrated around 20–25 keV.

The SpekPy comparison for this configuration is reported in Figure 4. As at 150 keV, the two codes agree on the overall shape, while the same systematic differences highlighted in Section 2 (binning of the histogram and low-energy tail) are visible here. At 28 keV the spectrum is dominated by a narrow filtered bremsstrahlung continuum with no characteristic peaks, since E_0 lies below the K-edge of the target.

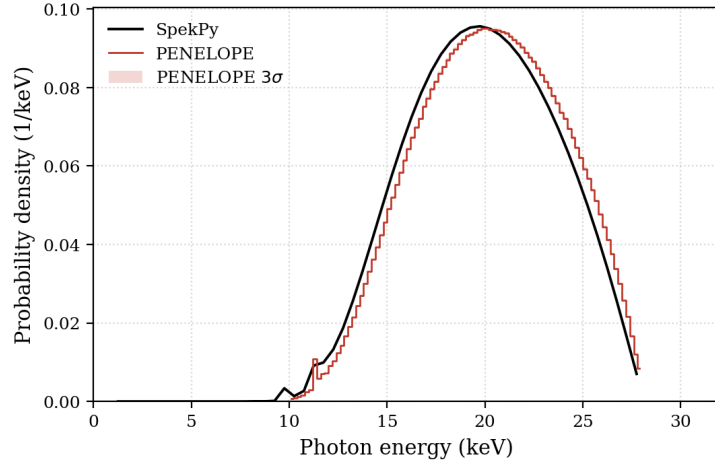


Figure 4: Comparison between PENELOPE (red, 3σ band) and SpekPy v2 (black) for the W, 28 keV, 45° , 1 mm Al case.

4. Effect of Anode Material ($W \rightarrow Mo$ at 28 keV)

Replacing the W anode with Mo (Figure 5) at the same 28 keV beam energy and geometry has a dramatic effect: the Mo characteristic K-lines now appear in the spectrum, because the K-shell binding energy of Mo (~ 20.0 keV) is below $E_0 = 28$ keV and the K-shell can be efficiently ionised.

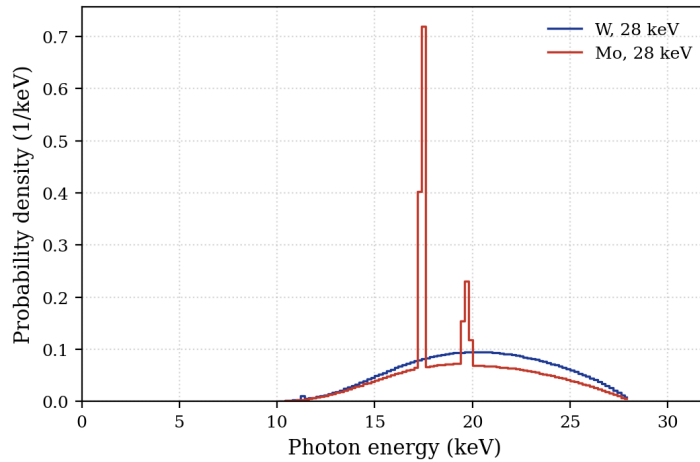


Figure 5: Effect of replacing the anode material at fixed energy and geometry (28 keV, 45° , 1 mm Al). The Mo K-lines at ~ 17.5 keV ($K\alpha$) and ~ 19.6 keV ($K\beta$) dominate the spectrum.

The two visible lines are the $K\alpha$ doublet at ~ 17.5 keV and the $K\beta$ line at ~ 19.6 keV. These peaks are very prominent because, with E_0 only slightly above the K-edge, K-shell ionisation absorbs a large fraction of the electron kinetic energy and the bremsstrahlung continuum is correspondingly weak. The resulting spectrum is concentrated in a narrow window in the 17–20 keV region, with a much sharper structure than the W case at the same beam energy.

Figure 6 shows the SpekPy comparison for this case. The analytical model reproduces the position of the Mo $K\alpha$ and $K\beta$ peaks correctly; the relative intensity of the lines and the bremsstrahlung pedestal are also in good agreement, well within the combined statistical and binning uncertainty.

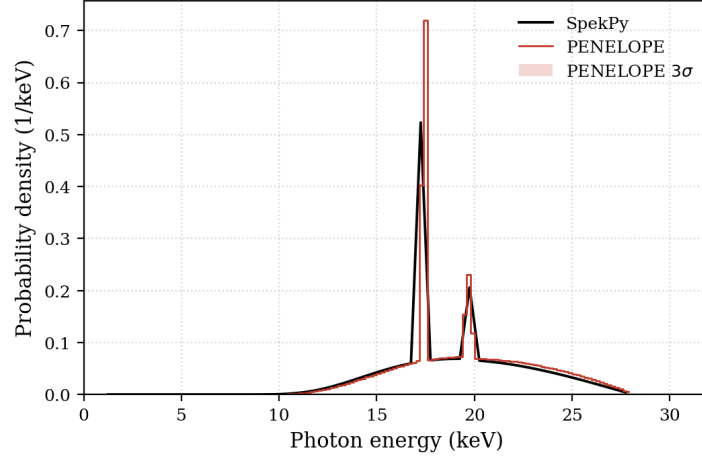


Figure 6: Comparison between PENELOPE (red, 3σ band) and SpekPy v2 (black) for the Mo, 28 keV, 45° , 1 mm Al case.

5. Effect of Anode Angle and Filtration

Two distinct effects are at play when going from $45^\circ/1$ mm to $20^\circ/2$ mm.

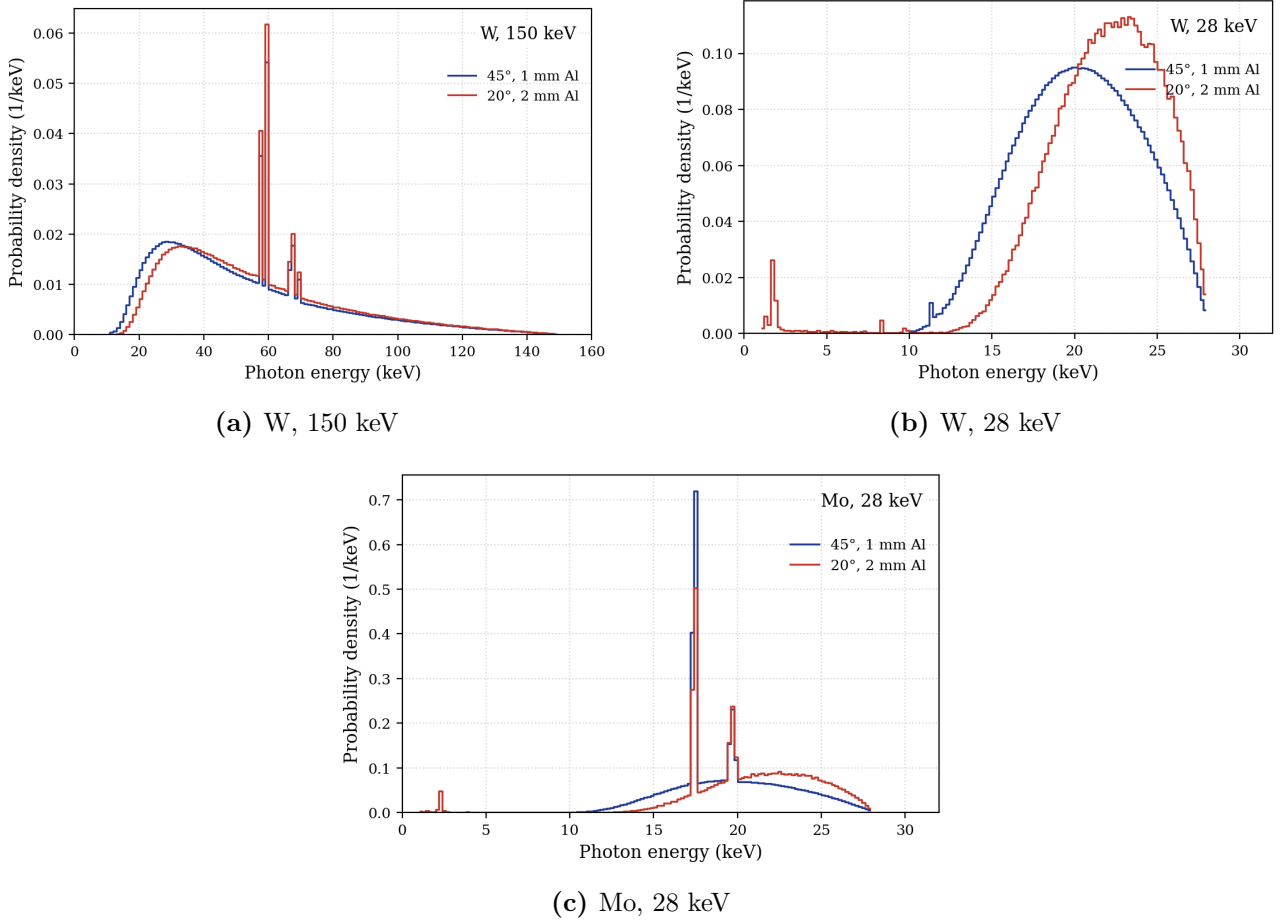


Figure 7: Comparison of the three configurations between the original geometry (45° anode, 1 mm Al, blue) and the modified one (20° anode, 2 mm Al, red).

The reduction of the anode tilt from 45° to 20° increases the path length that each photon must travel through the tungsten or molybdenum target before exiting; this anode self-

attenuation preferentially absorbs low-energy photons, shifting the spectrum towards higher energies. At the same time, doubling the Al thickness from 1 mm to 2 mm reinforces the same trend, since the linear attenuation coefficient of Al falls rapidly with energy and removes the soft photons more aggressively.

The net effect (Figure 7) is a beam-hardening of all three spectra: their mean energy increases and their low-energy tail is suppressed. The magnitude of the effect is, however, strongly energy-dependent, and is discussed case by case below.

Tungsten at 150 keV. At the diagnostic energy the change is mild and visible only as a small shift of the bremsstrahlung peak, while the K-line region is unchanged. The SpekPy comparison for the modified geometry, shown in Figure 8, reproduces this small hardening faithfully.

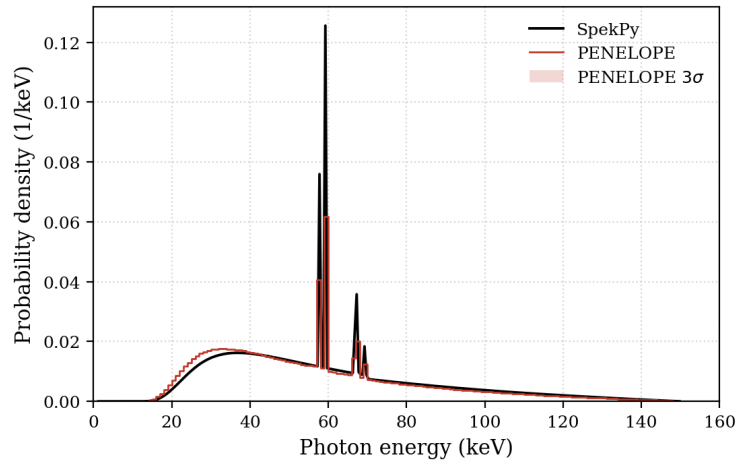


Figure 8: Comparison between PENELOPE (red, 3σ band) and SpekPy v2 (black) for the W, 150 keV, 20° , 2 mm Al case.

Tungsten at 28 keV. At the mammographic energy the impact is much larger because the Al attenuation coefficient is significantly higher in this range: the peak of the bremsstrahlung continuum shifts visibly to higher energies and the spectrum becomes noticeably narrower. Figure 9 shows the corresponding SpekPy overlay; the analytical model reproduces both the position and the width of the hardened continuum.

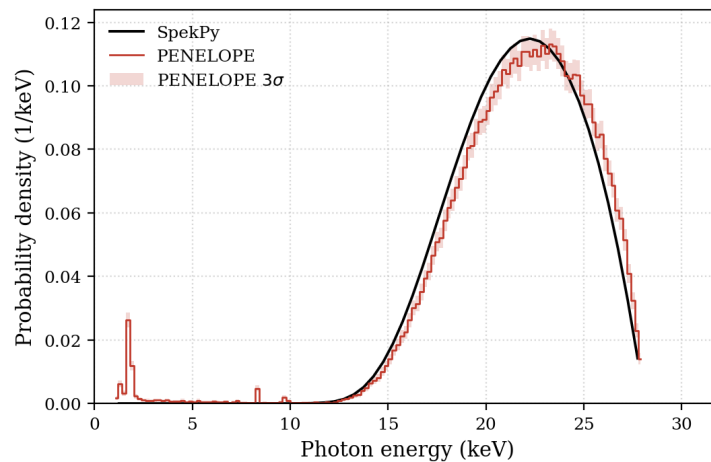


Figure 9: Comparison between PENELOPE (red, 3σ band) and SpekPy v2 (black) for the W, 28 keV, 20° , 2 mm Al case.

Molybdenum at 28 keV. For the Mo target the bremsstrahlung background is further reduced relative to the K-lines, making the spectrum even cleaner around the $K\alpha/K\beta$ doublet. The SpekPy comparison (Figure 10) is consistent with PENELOPE: the relative intensity of the K-lines and the surviving continuum agree within the statistical and binning uncertainty.

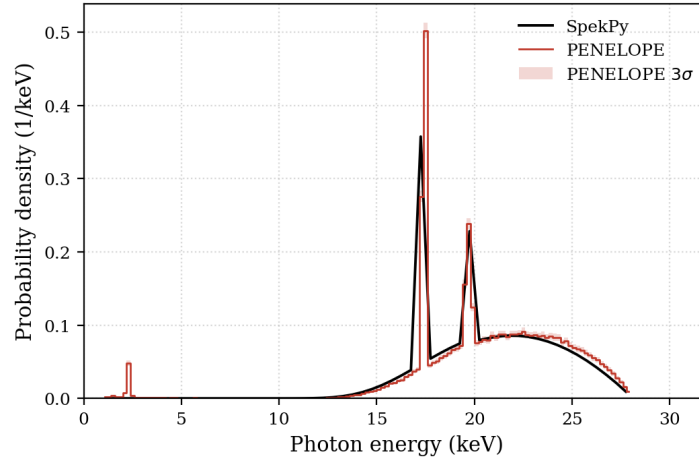


Figure 10: Comparison between PENELOPE (red, 3σ band) and SpekPy v2 (black) for the Mo, 28 keV, 20° , 2 mm Al case.

The combined geometry/filtration change therefore always hardens the beam, but its impact is much more pronounced at the mammographic energy of 28 keV than at 150 keV, because the relative weight of Al filtration is dominant where the photon energies are low. The same kind of small systematic differences described in Section 2 (peak height and low-energy tail) is observed in all three SpekPy comparisons — no new sources of disagreement appear when the geometry is changed.

6. Conclusions

The six configurations explored in this practice illustrate, in sequence, the three main levers available to the designer of an X-ray tube spectrum: *electron beam energy*, *anode material*, and *geometry/filtration*. The endpoint of the spectrum is fixed by E_0 , since no photon can be emitted with an energy higher than the kinetic energy of the impacting electron. The position of the characteristic lines is fixed by the anode material (W K-lines at 58–70 keV, Mo K-lines at 17–20 keV) and is observed only when E_0 is high enough to ionise the corresponding K shell, which is why the W K-lines disappear at the mammographic energy of 28 keV while the Mo K-lines emerge prominently. Finally, the shape of the continuum below the endpoint is moulded by the inherent self-attenuation in the anode and by the added Al filter, both of which preferentially remove low-energy photons and harden the beam — an effect that is much stronger at 28 keV than at 150 keV because the Al attenuation coefficient grows steeply at low photon energies.

The PENELOPE Monte Carlo results are in good agreement with the analytical SpekPy prediction across all six configurations, both in the diagnostic and in the mammographic energy range. The residual differences between the two codes (small mismatches in K-line peak height, histogram binning, and behaviour close to the transport cut-off) are systematic, do not depend on the configuration, and remain within the statistical uncertainty of the simulation.